

Series, Approximation, Orders of Magnitude

How physicists actually think — and why every theory in this book is the first few terms of another one.

WHERE WE ARE

Here is the thesis, stated as hard as it can be stated: **physics is almost never solved exactly; it is expanded.** The list of physically interesting systems with closed-form solutions is embarrassingly short — two bodies under gravity, the harmonic oscillator, hydrogen, a handful of others. Everything else you have ever read about, from planetary orbits to the electron's magnetic moment, was obtained by identifying a small parameter, expanding in it, and keeping the terms that mattered.

Chapter 0.1 gave you the first term of such an expansion and noticed, almost in passing, that the leftover error was *quadratic*. This chapter keeps going. It turns that observation into a theorem with an error bound, and then into the single most productive habit in theoretical physics: *find the small parameter, expand in it, keep what you need, and know what you threw away.*

Everything in this book that sounds like a slogan — "Newton is the low-speed limit of Einstein", "classical mechanics is the $\hbar \rightarrow 0$ limit of quantum mechanics", "the Standard Model is an effective theory" — is, literally and unromantically, a statement about the first surviving term of a series. By the end of this chapter you will be able to read those sentences as mathematics rather than as reassurance.

TOOLS YOU'LL NEED — Chapter 0.1: linearization, higher derivatives, the definition of e . Chapter 0.2: the fundamental theorem of calculus and integration by parts, which is the engine of §1.

1 · Taylor's theorem, derived

Almost every book states Taylor's series and then, some pages later, mumbles about a remainder. We will do it the other way round, because the remainder is the entire point. A series without an error estimate is not a result; it is a wish.

Start from the fundamental theorem of calculus, which says nothing more than "the function at the end equals the function at the start plus the accumulated change":

$$f(a + h) = f(a) + \int_a^{a+h} f'(t) dt. \quad (0.3.1)$$

That is *exact*. No approximation has been made and none will be; every step below is an identity. Now integrate by parts. The trick — and it is the only trick in the entire derivation — is the choice of antiderivative. Integrating dt gives $t + C$ for any constant C , and we choose the constant to make the upper boundary term vanish: take the antiderivative to be $t - (a + h)$, which is zero at $t = a + h$. Then

$$\int_a^{a+h} \underbrace{f'(t)}_u \underbrace{dt}_{dv} = \left[f'(t) (t - (a + h)) \right]_a^{a+h} - \int_a^{a+h} f''(t) (t - (a + h)) dt. \quad (0.3.2)$$

The boundary term at $t = a + h$ is zero by construction; at $t = a$ it is $-f'(a)(a - a - h) = +f'(a)h$, and it enters with a minus sign in front of the bracket's lower limit, so it contributes $+f'(a)h$. Flipping the sign inside the remaining integral,

$$f(a + h) = f(a) + f'(a)h + \int_a^{a+h} (a + h - t) f''(t) dt. \quad (0.3.3)$$

Stop and look at what you have. The first two terms are exactly the linear approximation of Chapter 0.1 — and the vague phrase "smaller than h " has been replaced by an explicit integral. That integral is the error, written down in full, with no limits taken and nothing swept anywhere.

Now do it again. In $\int (a + h - t) f''(t) dt$, let $u = f''(t)$ and $dv = (a + h - t) dt$, whose antiderivative $-\frac{1}{2}(a + h - t)^2$ again vanishes at the upper limit:

$$\begin{aligned} \int_a^{a+h} (a + h - t) f''(t) dt &= \underbrace{\left[-\frac{1}{2}(a + h - t)^2 f''(t) \right]_a^{a+h}}_{= +\frac{h^2}{2} f''(a)} \\ &\quad + \int_a^{a+h} \frac{(a + h - t)^2}{2} f'''(t) dt. \end{aligned} \quad (0.3.4)$$

So

$$f(a + h) = f(a) + f'(a)h + \frac{1}{2}f''(a)h^2 + \int_a^{a+h} \frac{(a + h - t)^2}{2!} f'''(t) dt. \quad (0.3.5)$$

There is the $\frac{1}{2}f''(a)h^2$ that the zoom figure in Chapter 0.1 measured numerically. It was never a coincidence; it is the second boundary term. And the pattern is now visible: each integration by parts peels off one more term of a polynomial and hands back an integral with one more power of $(a + h - t)$ and one more derivative on f . Doing it n times gives **Taylor's theorem with the integral form of the remainder**:

$$\begin{aligned}
f(a+h) &= \sum_{k=0}^n \frac{f^{(k)}(a)}{k!} h^k + R_{n+1}, \\
R_{n+1} &= \int_a^{a+h} \frac{(a+h-t)^n}{n!} f^{(n+1)}(t) dt.
\end{aligned}
\tag{0.3.6}$$

(As usual $f^{(0)} = f$ and $0! = 1$, so the $k = 0$ term is just $f(a)$.) Setting $a = 0$ and renaming $h \rightarrow x$ gives the **Maclaurin** form, which is what you will use nine times out of ten: $f(x) = \sum_k f^{(k)}(0)x^k/k! + R$.

▼ Grind box — the induction, and the Lagrange form of the remainder

The induction. Define, for $n \geq 1$,

$$R_n \equiv \int_a^{a+h} \frac{(a+h-t)^{n-1}}{(n-1)!} f^{(n)}(t) dt.$$

Equation (0.3.1) says $f(a+h) = f(a) + R_1$, which starts the induction. Integrating by parts with $u = f^{(n)}(t)$ and antiderivative $v = -\frac{(a+h-t)^n}{n!}$ (again chosen to vanish at $t = a+h$):

$$R_n = \left[-\frac{(a+h-t)^n}{n!} f^{(n)}(t) \right]_a^{a+h} + \int_a^{a+h} \frac{(a+h-t)^n}{n!} f^{(n+1)}(t) dt = \frac{h^n}{n!} f^{(n)}(a) + R_{n+1}.$$

Each application converts a remainder into "the next polynomial term plus the next remainder", so applying it n times to R_1 gives (0.3.6) exactly. The only hypothesis used is that $f^{(n+1)}$ exists and is continuous on the closed interval — enough for integration by parts to be legal.

Lagrange form. The integral remainder is exact but not always easy to look at. Note first that, substituting $s = a+h-t$,

$$\int_a^{a+h} \frac{(a+h-t)^n}{n!} dt = \int_0^h \frac{s^n}{n!} ds = \frac{h^{n+1}}{(n+1)!}.$$

For $h > 0$ the weight $(a+h-t)^n/n!$ is non-negative on the whole interval, so the mean value theorem for integrals applies: there is some ξ between a and $a+h$ with

$$R_{n+1} = f^{(n+1)}(\xi) \int_a^{a+h} \frac{(a+h-t)^n}{n!} dt = \frac{f^{(n+1)}(\xi)}{(n+1)!} h^{n+1}.$$

(For $h < 0$ reverse the orientation of the interval; the weight is then of one sign again and the same conclusion follows.) This is the form you will actually use for estimates, because you rarely know ξ but you can almost always *bound* $|f^{(n+1)}|$ on the interval.

WHAT THE REMAINDER IS FOR

Read the Lagrange form as a sentence: **the error you make by truncating after the h^n term is itself a Taylor term — the very next one — with the derivative evaluated somewhere you don't know.** That single fact is why physicists can be so casual and so accurate at the same time. You do not need ξ ; you need a bound on the next derivative, and then

$$|R_{n+1}| \leq \frac{\max |f^{(n+1)}|}{(n+1)!} |h|^{n+1}.$$

Two things control the error: how big the next derivative can get, and how fast $|h|^{n+1}$ falls. The factorial in the denominator is the reason a handful of terms is usually enough — and in §4 we will meet the case where the derivatives grow *faster* than the factorial, which is where physics actually lives.

One warning to carry into §3, since it costs nothing to state now. The Taylor *series* is what you get by letting $n \rightarrow \infty$ in (0.3.6). It equals f precisely when $R_{n+1} \rightarrow 0$ — *not* merely when the series converges. Those are different conditions, and the difference is the subject of a callout later in this chapter that you may find genuinely unsettling.

2 · The essential expansions

Five expansions carry most of the load in physics. We derive each one, because the derivations take one line apiece and because you should never be in the position of half-remembering whether there is a factorial in the binomial series.

The exponential

By definition (Chapter 0.1) e^x is its own derivative, so $f^{(k)}(x) = e^x$ and $f^{(k)}(0) = 1$ for every k . Feeding that into the Maclaurin form:

$$e^x = \sum_{k=0}^{\infty} \frac{x^k}{k!} = 1 + x + \frac{x^2}{2} + \frac{x^3}{6} + \frac{x^4}{24} + \cdots \quad (0.3.7)$$

Does it converge to e^x ? Use the Lagrange bound. On the interval between 0 and x we have $e^\xi \leq e^{|x|}$, so $|R_{n+1}| \leq e^{|x|} |x|^{n+1} / (n+1)!$. Once $n+1 > 2|x|$, every further step multiplies this by $|x| / (n+2) < \frac{1}{2}$, so the bound is eventually smaller than a constant times 2^{-n} and therefore goes to zero. The series converges to the function for *every* real x — and, as we are about to exploit, for every complex x too.

Sine and cosine

Derivatives of sin cycle with period four: sin, cos, $-\sin$, $-\cos$, $\dots$, which at $x = 0$ evaluate to 0, 1, 0, -1 , $\dots$. Only odd powers survive, alternating in sign:

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots, \quad \cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \cdots, \quad (0.3.8)$$

the second by the identical argument starting from $\cos$ (values $1, 0, -1, 0, \dots$ at the origin). Convergence everywhere is immediate: every derivative of $\sin$ or $\cos$ is bounded by 1 in absolute value, so $|R_{n+1}| \leq |x|^{n+1} / (n+1)! \rightarrow 0$ by the same factorial argument.

Note in passing that the leading term of $\sin x$ is x , which is the "small-angle approximation" you have used since school — and that the first correction is $-x^3/6$, which tells you exactly when to stop using it. At $x = 0.1$ rad the correction is 1.7×10^{-4} , a relative error of 0.17%. That number is not a rule of thumb; it is a computed quantity.

The logarithm

Differentiating $\ln(1+x)$ repeatedly gives $f' = (1+x)^{-1}$, $f'' = -(1+x)^{-2}$, $f''' = 2(1+x)^{-3}$, and in general $f^{(k)}(x) = (-1)^{k-1}(k-1)!(1+x)^{-k}$. At $x = 0$ that is $(-1)^{k-1}(k-1)!$, and dividing by $k!$ leaves $(-1)^{k-1}/k$:

$$\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \cdots \quad (|x| < 1). \quad (0.3.9)$$

The restriction $|x| < 1$ is not decoration and §3 explains where it comes from. The leading behaviour $\ln(1+x) \approx x$ is one of the two most-used approximations in quantitative science — it is why small log-hazard-ratios read directly as percentages, a point we return to in the Familiar Ground box.

The binomial series, for arbitrary real exponent

This is the workhorse. Let $f(x) = (1+x)^\alpha$ with α any real number — not necessarily a positive integer. Then $f' = \alpha(1+x)^{\alpha-1}$, $f'' = \alpha(\alpha-1)(1+x)^{\alpha-2}$, and in general

$$\begin{aligned} f^{(k)}(x) &= \alpha(\alpha-1) \cdots (\alpha-k+1) (1+x)^{\alpha-k} \\ \implies f^{(k)}(0) &= \alpha(\alpha-1) \cdots (\alpha-k+1). \end{aligned} \quad (0.3.10)$$

Dividing by $k!$ defines the generalised binomial coefficient, and

$$(1+x)^\alpha = \sum_{k=0}^{\infty} \binom{\alpha}{k} x^k, \quad \binom{\alpha}{k} \equiv \frac{\alpha(\alpha-1) \cdots (\alpha-k+1)}{k!}, \quad |x| < 1. \quad (0.3.11)$$

When α is a non-negative integer the product eventually hits a factor of zero, the series terminates, and you recover the finite binomial theorem you already know. For every other α it runs forever. Two cases deserve to be committed to memory, because between them they cover essentially every square root you will ever have to linearize:

$$\begin{aligned}\sqrt{1+x} &= 1 + \frac{x}{2} - \frac{x^2}{8} + \frac{x^3}{16} - \dots, \\ \frac{1}{\sqrt{1-x}} &= 1 + \frac{x}{2} + \frac{3x^2}{8} + \frac{5x^3}{16} + \dots\end{aligned}\tag{0.3.12}$$

The second is $\alpha = -\frac{1}{2}$ with $x \rightarrow -x$, and it is the relativistic γ factor. Put $x = v^2/c^2$ and you have, in one line, the expansion that Chapter 0.1 obtained by grinding out two derivatives — and now you get every higher term for free, which is exactly what Worked Example 2 will cash in.

▼ **Grind box — where the radius $|x| < 1$ comes from, without complex analysis**

For the logarithm you can bypass Taylor's theorem entirely and get an exact remainder from the finite geometric sum. For any $t \neq -1$ and any n , multiply out to verify

$$\frac{1}{1+t} = 1 - t + t^2 - \dots + (-t)^{n-1} + \frac{(-t)^n}{1+t}.$$

This is an algebraic identity, not an approximation: the finite sum telescopes when you multiply both sides by $(1+t)$. Now integrate from 0 to x , using $\int_0^x t^k dt = x^{k+1}/(k+1)$ and $\int_0^x dt/(1+t) = \ln(1+x)$:

$$\ln(1+x) = x - \frac{x^2}{2} + \dots + \frac{(-1)^{n-1}x^n}{n} + \underbrace{(-1)^n \int_0^x \frac{t^n}{1+t} dt}_{R_n}.$$

For $0 \leq x < 1$ the integrand is at most t^n , so $|R_n| \leq x^{n+1}/(n+1) \rightarrow 0$ and the series converges to the function. For $x > 1$ the terms x^n/n themselves blow up, so the series cannot converge at all: a convergent series must have terms tending to zero. The radius is exactly 1, and the reason is visible on the real line — $\ln(1+x)$ runs to $-\infty$ at $x = -1$.

The binomial radius. Take the ratio of consecutive coefficients in (0.3.11):

$$\left| \frac{\binom{\alpha}{k+1} x^{k+1}}{\binom{\alpha}{k} x^k} \right| = \left| \frac{\alpha - k}{k+1} \right| |x| \longrightarrow |x| \quad \text{as } k \rightarrow \infty.$$

By the ratio test of §3 the series converges for $|x| < 1$ and diverges for $|x| > 1$, for every non-integer α . Again the reason is visible: $(1+x)^\alpha$ misbehaves at $x = -1$, one unit from the origin.

Reference table

FUNCTION	SERIES ABOUT 0	RADIUS	WHERE IT IS SPENT IN THIS BOOK
e^x	$1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots$	∞	Everywhere. Time evolution, partition functions, propagators
$\sin x$	$x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots$	∞	Small oscillations (0.8), string modes (7.4)
$\cos x$	$1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \dots$	∞	Pendulum potential (below), lattice dispersion
$\ln(1+x)$	$x - \frac{x^2}{2} + \frac{x^3}{3} - \dots$	1	Entropy, log-likelihoods, running couplings (5.9)
$\frac{1}{1-x}$	$1 + x + x^2 + x^3 + \dots$	1	Geometric resummation; the propagator in 5.6
$(1+x)^\alpha$	$1 + \alpha x + \frac{\alpha(\alpha-1)}{2}x^2 + \dots$	1	Every square root
$\sqrt{1+x}$	$1 + \frac{x}{2} - \frac{x^2}{8} + \frac{x^3}{16} - \dots$	1	$E = \sqrt{p^2 c^2 + m^2 c^4}$ (2.5)
$(1-x)^{-1/2}$	$1 + \frac{x}{2} + \frac{3x^2}{8} + \frac{5x^3}{16} + \dots$	1	The Lorentz factor γ (2.5)

Euler's formula, and why complex numbers are not a convenience

The exponential series (0.3.7) converges for every real x , and its convergence proof used only $|x|$ — so it converges just as well for complex arguments. We therefore *define* e^z for complex z by that series; there is no other sensible choice, since any definition agreeing with e^x on the real line and having a power series must be this one.

Now substitute $z = i\theta$ with θ real, and use the four-fold cycle $i^0 = 1$, $i^1 = i$, $i^2 = -1$, $i^3 = -i$, which then repeats:

$$e^{i\theta} = \sum_{k=0}^{\infty} \frac{(i\theta)^k}{k!} = \underbrace{1 - \frac{\theta^2}{2!} + \frac{\theta^4}{4!} - \dots}_{\text{even } k, i^{2m}=(-1)^m} + i \underbrace{\left(\theta - \frac{\theta^3}{3!} + \frac{\theta^5}{5!} - \dots \right)}_{\text{odd } k, i^{2m+1}=i(-1)^m}. \tag{0.3.13}$$

Splitting the sum into even and odd k is legitimate because the series converges *absolutely* — the sum of $|\theta|^k/k!$ converges — and absolutely convergent series may be rearranged freely without changing their value. Compare the two brackets with (0.3.8) and read off

$$e^{i\theta} = \cos \theta + i \sin \theta$$

(0.3.14)

which is not an identity between three unrelated functions but a statement that they were the same object all along, sorted by parity. Three consequences, each of which we will use repeatedly.

1. It has unit modulus. $|e^{i\theta}|^2 = \cos^2 \theta + \sin^2 \theta = 1$. So $e^{i\theta}$ is a point on the unit circle at angle θ , and θ is arc length along that circle. The identity $e^{i\pi} = -1$ is the special case "walk half way round".

2. Multiplying by $e^{i\theta}$ is a rotation. Take any complex number $z = x + iy$, thought of as the point (x, y) in the plane, and multiply:

$$z e^{i\theta} = (x + iy)(\cos \theta + i \sin \theta) = \underbrace{(x \cos \theta - y \sin \theta)}_{\text{new } x} + i \underbrace{(x \sin \theta + y \cos \theta)}_{\text{new } y}. \quad (0.3.15)$$

Those are precisely the components of the two-dimensional rotation matrix acting on (x, y) . Rotation, one of the central objects of the rest of this book, is multiplication by a complex exponential. In Chapter 0.4 this becomes the statement that $U(1)$ and $SO(2)$ are the same group, and in Chapter 6.3 the fact that the phase of a quantum field lives on a circle is what generates electromagnetism.

3. It converts decay into oscillation. Chapter 0.1 left you holding a comparison: the clearance equation $\dot{C} = -kC$ gives $C = C_0 e^{-kt}$, while quantum time evolution $i\hbar \partial_t \psi = E\psi$ gives $\psi = \psi_0 e^{-iEt/\hbar}$. Structurally identical; the only difference is the i . Euler's formula now tells you exactly what that i does:

$$\begin{aligned} e^{-kt} &\longrightarrow \text{shrinks towards } 0, \\ e^{-iEt/\hbar} &= \cos \frac{Et}{\hbar} - i \sin \frac{Et}{\hbar} \longrightarrow \text{rotates, at constant modulus } 1. \end{aligned} \quad (0.3.16)$$

A decaying exponential loses amplitude; an imaginary exponential keeps its length and changes only its direction. That is the whole reason quantum mechanics conserves probability: $|\psi|^2$ is the squared length of a vector that is being *rotated*, and rotations preserve length. This is also why complex numbers are not an accounting convenience in quantum mechanics. Real exponentials can only grow or decay; you need the circle, and the circle is $e^{i\theta}$.

One free bonus, to show the machinery paying rent immediately. From $e^{i(\alpha+\beta)} = e^{i\alpha}e^{i\beta}$, expand both sides with (0.3.14) and equate real and imaginary parts:

$$\begin{aligned} \cos(\alpha + \beta) + i \sin(\alpha + \beta) &= (\cos \alpha + i \sin \alpha)(\cos \beta + i \sin \beta) \\ \implies \cos(\alpha + \beta) &= \cos \alpha \cos \beta - \sin \alpha \sin \beta, \\ \sin(\alpha + \beta) &= \sin \alpha \cos \beta + \cos \alpha \sin \beta. \end{aligned} \quad (0.3.17)$$

The trigonometric identities you were made to memorise are the statement that $e^{i\theta}$ obeys the exponential law. There is nothing else to them.

3 · Convergence, radius, and a genuine surprise

A power series $\sum_k a_k x^k$ is an infinite sum, and infinite sums are promises. We need a test for when the promise is kept.

The ratio test

Suppose $|c_{k+1}/c_k| \rightarrow L$. If $L < 1$, pick any r with $L < r < 1$. Then beyond some index K we have $|c_{k+1}| \leq r |c_k|$, hence $|c_{K+m}| \leq r^m |c_K|$. The tail of the series is therefore dominated term by term by the geometric series $|c_K| \sum_m r^m = |c_K|/(1-r)$, which is finite; a series of positive terms with bounded partial sums converges, because its partial sums increase and are bounded above. If instead $L > 1$ the terms eventually grow, so they cannot tend to zero, and a series whose terms do not tend to zero cannot converge. Hence:

$$\begin{aligned} \lim_{k \rightarrow \infty} \left| \frac{c_{k+1}}{c_k} \right| < 1 &\implies \sum_k c_k \text{ converges absolutely,} \\ \lim_{k \rightarrow \infty} \left| \frac{c_{k+1}}{c_k} \right| > 1 &\implies \sum_k c_k \text{ diverges.} \end{aligned} \tag{0.3.18}$$

Apply this to a power series by putting $c_k = a_k x^k$. The ratio is $|a_{k+1}/a_k| |x|$, so if $|a_k/a_{k+1}| \rightarrow R$ the series converges for $|x| < R$ and diverges for $|x| > R$. That number R is the **radius of convergence**. For e^x , $|a_k/a_{k+1}| = (k+1)/k! \rightarrow \infty$: the radius is infinite. For $\ln(1+x)$, $|a_k/a_{k+1}| = (k+1)/k \rightarrow 1$: the radius is 1, as the grind box already showed by hand.

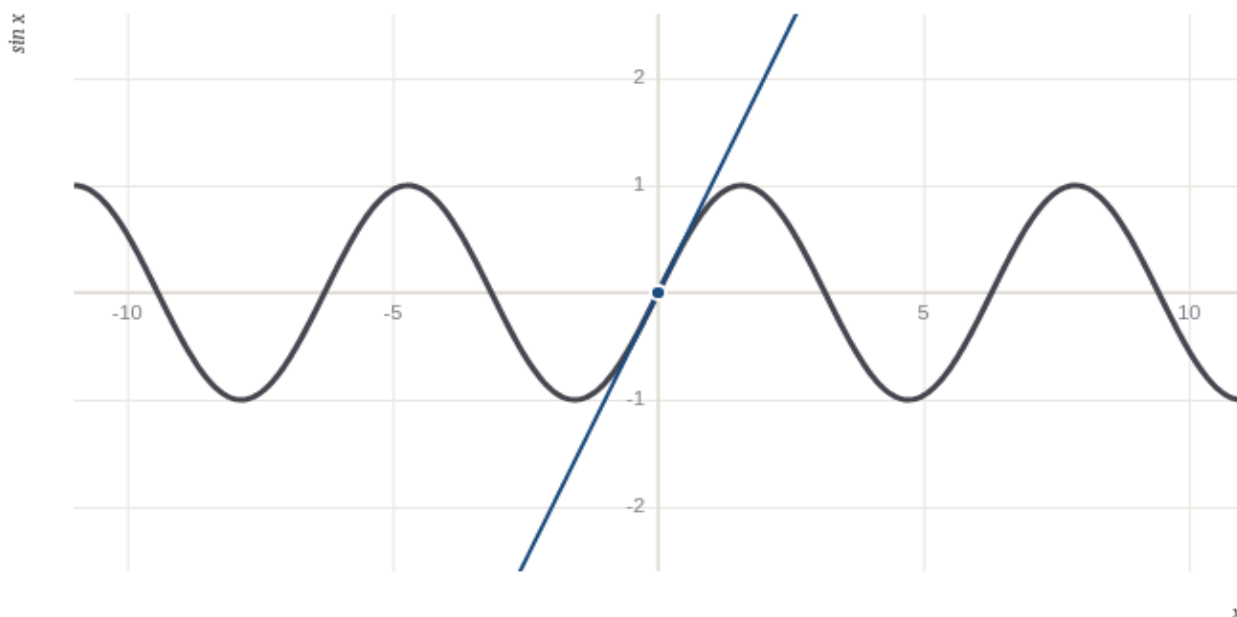
The surprise

Here is a function with nothing whatsoever wrong with it:

$$f(x) = \frac{1}{1+x^2} = 1 - x^2 + x^4 - x^6 + \dots \tag{0.3.19}$$

(the series follows from $\frac{1}{1-u} = \sum u^k$ with $u = -x^2$, or from the same exact finite-sum identity used in the grind box). On the real line f is smooth, positive, bounded by 1, infinitely differentiable, and utterly featureless — it is the Lorentzian lineshape, and it does nothing interesting at $x = \pm 1$ or anywhere else. Yet by the ratio test its series converges only for $|x| < 1$. At $x = 2$ the terms are $1, -4, 16, -64, \dots$; the "approximation" runs away to infinity no matter how many terms you take.

Something is setting a radius of 1, and it is not visible on the real line. Watch it happen:



target sin x $1/(1+x^2)$ order N $N = 1$ trust radius ≈ 0.49

Two fates. Grey is the true function, blue is the Taylor polynomial of order N about the origin. The readout reports the *trust radius*: how far from the origin the polynomial stays within 0.02 of the truth. For $\sin x$ the trust radius marches outward without limit as you add terms — the series converges everywhere. Switch to $1/(1+x^2)$ and the trust radius stalls dead at $|x| = 1$ (orange lines) and refuses to move, while outside that boundary each extra term makes the approximation *worse*. Nothing happens to the function at $|x| = 1$. Something is happening somewhere else.

Where the boundary actually is

The series in (0.3.19) makes perfect sense with a complex variable z in place of x , and there the answer is immediate: $1/(1+z^2)$ blows up at $z = \pm i$, both at distance 1 from the origin. In fact at $z = i$ every term of the series equals $(-1)^k i^{2k} = (-1)^k (-1)^k = 1$, so the partial sums are $1, 2, 3, \dots$ — the series does not merely fail to represent the function there, it diverges outright.

That is enough to pin the radius, given one small lemma: the set where a power series converges is a *disc*. If $\sum a_k z_1^k$ converges then its terms are bounded, say $|a_k z_1^k| \leq M$, and for any z with $|z| < |z_1|$,

$$|a_k z^k| = |a_k z_1^k| \left| \frac{z}{z_1} \right|^k \leq M \left| \frac{z}{z_1} \right|^k, \quad (0.3.20)$$

which is a convergent geometric series. So convergence at one point forces absolute convergence everywhere strictly closer to the origin. Contrapositive: divergence at $z = i$ forces divergence at every $|z| > 1$. The radius is exactly 1, and it was determined by two points that do not lie on the real line at all.

THE MORAL, AND IT IS A DEEP ONE

The real line does not know why its own series fails; the complex plane does. A real function can look flawless over its entire domain and still have a Taylor series that gives up at a specific radius, because the radius is set by the nearest singularity *in the complex plane* — a place your real-valued function never visits and never reports on.

This is not a curiosity. It is the reason that analyticity, poles and branch cuts become physical objects later in the book. In Chapter 5.9 a scattering amplitude is treated as an analytic function of complex energy, and then: poles on the real axis are stable particles, poles just off it are resonances with lifetimes given by the imaginary part, and branch cuts are thresholds where new particles can be produced. The analytic structure in a region you cannot experimentally visit controls what you measure in the region you can. That entire subject is this footnote about $1/(1+x^2)$, taken seriously.

4 · Asymptotic series — where physics really lives

Everything so far has been about series that converge. Now the uncomfortable truth: most of the series physics actually runs on do not converge, for any value of the parameter, ever. They are still the most accurate predictions humanity has made. That apparent contradiction is resolved by a single idea, and this section is where the chapter earns its title.

Two ways to take a limit

A power series statement involves two variables, the argument x and the number of terms N , and there are two different things you can mean.

Convergent: fix x , let $N \rightarrow \infty$, and demand the partial sums approach $f(x)$. That is what §3 was about.

Asymptotic: fix N , let $x \rightarrow 0$, and demand the error die faster than the last term you kept. Formally, $f(x) \sim \sum_k a_k x^k$ as $x \rightarrow 0^+$ means that for each fixed N ,

$$\lim_{x \rightarrow 0^+} \frac{f(x) - \sum_{k=0}^N a_k x^k}{x^N} = 0. \quad (0.3.21)$$

These are genuinely different demands, and **the two limits do not commute**. A series can satisfy (0.3.21) for every N while diverging for every $x \neq 0$. The convergent case asks "does adding terms help forever?"; the asymptotic case asks "does the approximation improve as the parameter gets small?". Physics almost always wants the second question, because in physics the parameter is small and given to you — v/c , $\hbar$, the fine structure constant — and the number of terms is a choice you make.

A completely explicit example

Consider

$$F(x) = \int_0^\infty \frac{e^{-t}}{1+xt} dt, \quad x > 0. \quad (0.3.22)$$

This integral is perfectly well defined and finite for every $x > 0$: the integrand is positive and bounded by e^{-t} , so $0 < F(x) \leq 1$. Now expand the denominator with the exact finite-sum identity from the grind box, $\frac{1}{1+u} = \sum_{k=0}^N (-u)^k + \frac{(-u)^{N+1}}{1+u}$ with $u = xt$, and integrate term by term using $\int_0^\infty e^{-t} t^k dt = k!$ (integrate by parts k times):

$$\begin{aligned} F(x) &= \sum_{k=0}^N (-1)^k k! x^k + R_{N+1}(x), \\ R_{N+1}(x) &= (-x)^{N+1} \int_0^\infty \frac{e^{-t} t^{N+1}}{1+xt} dt. \end{aligned} \quad (0.3.23)$$

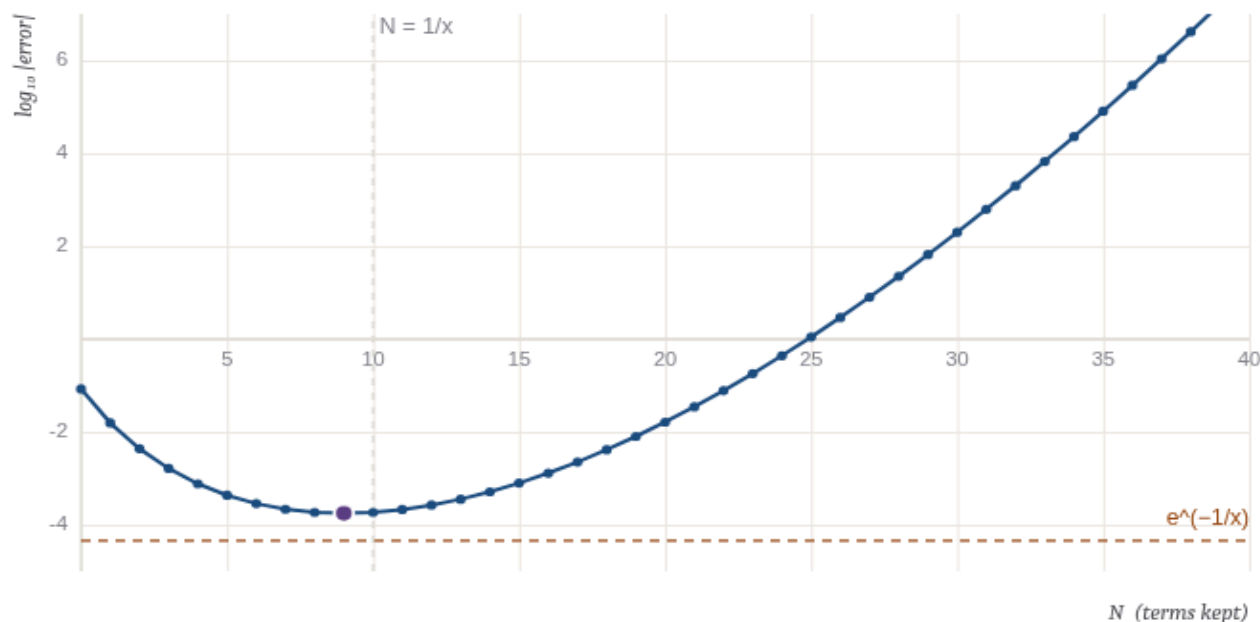
Everything here is exact. And since $1/(1+xt) \leq 1$ for $x, t > 0$, the remainder obeys

$$|R_{N+1}(x)| \leq x^{N+1} \int_0^\infty e^{-t} t^{N+1} dt = (N+1)! x^{N+1}, \quad (0.3.24)$$

and the right-hand side is precisely the magnitude of the first term you dropped. That bound is beautiful, and it is the practical rule for asymptotic series: *the error is no bigger than the first omitted term*. It also proves (0.3.21) immediately, since $(N+1)! x^{N+1}/x^N = (N+1)! x \rightarrow 0$. So the series is asymptotic to F .

But apply the ratio test to $\sum (-1)^k k! x^k$: the ratio is $(k+1)|x|$, which tends to *infinity* for every $x \neq 0$. The radius of convergence is **zero**. The series diverges everywhere except the single point $x = 0$.

Both statements are true simultaneously, and the resolution is in the shape of the terms $k! x^k$. Multiply one term by $x(k+1)$ to get the next: while $k < 1/x$ that factor is less than one and the terms *shrink*; once $k > 1/x$ it exceeds one and they *grow*, forever. The series is useful up to the turning point and poisonous after it.



parameter x $x = 0.100$

best at $N = 9$, error = $1.77e-4$ ($e^{-1/x} = 4.54e-5$)

The dip and the explosion. Vertical axis is $\log_{10}$ of the actual error $|F(x) - S_N(x)|$, where S_N is the partial sum of the divergent series; horizontal axis is the number of terms kept. Adding terms helps — spectacularly — until $N \approx 1/x$ (dashed vertical), and then every further term makes things worse without bound. The dashed horizontal line is $e^{-1/x}$, the accuracy floor derived below. Slide x down and watch the floor drop through the decades while the useful range of N lengthens. This is what a divergent series looks like when it is being used correctly.

▼ Grind box — optimal truncation and the $e^{-1/x}$ accuracy floor

We want the N minimising the error bound $N! x^N$. Take logarithms and use the crude Stirling estimate, which we can derive on the spot by comparing a sum to an integral:

$$\ln N! = \sum_{k=1}^N \ln k \approx \int_1^N \ln t \, dt = N \ln N - N + 1.$$

(The comparison is legitimate because $\ln t$ is increasing, so the sum is trapped between $\int_0^N$ and $\int_1^{N+1}$; we need only the leading terms.) Then

$$\Phi(N) \equiv \ln(N! x^N) \approx N \ln N - N + N \ln x.$$

Differentiate with respect to N , treating it as continuous:

$$\Phi'(N) = \ln N + 1 - 1 + \ln x = \ln(Nx) = 0 \implies \boxed{N_* = \frac{1}{x}}.$$

It is a minimum because $\Phi'' = 1/N > 0$. Substituting back, at $N_* x = 1$:

$$\Phi(N_*) = N_* \ln(N_* x) - N_* = -N_* = -\frac{1}{x} \implies \text{best error} \sim e^{-1/x}.$$

Check it against the figure. At $x = 0.1$ this predicts stopping at about ten terms with an accuracy near $e^{-10} = 4.5 \times 10^{-5}$; the numerically computed optimum is $N = 9$ with error 1.8×10^{-4} . At $x = 0.05$: predicted floor $e^{-20} = 2.1 \times 10^{-9}$, actual optimum $N = 19$ with error 1.1×10^{-8} . The exponential scaling is exactly right; the factor of a few comes from the subleading $\sqrt{2\pi N}$ in Stirling that we discarded.

Remember this shape. An asymptotic series in a small parameter x delivers accuracy $e^{-1/x}$ and no better. That exponentially small leftover is not noise — it is a real physical effect that the power series is structurally incapable of seeing, as the warning box below explains.

The punchline

Now the statement this section exists for.

The perturbation series of quantum electrodynamics is asymptotic and divergent. It predicts the electron's magnetic moment to about twelve significant figures — the most accurately confirmed prediction in the history of science.

The expansion parameter is the fine structure constant $\alpha \approx 1/137$, and the series for the electron's anomalous magnetic moment $a_e = (g - 2)/2$ runs in powers of $\alpha/\pi \approx 1/430$. The measured value is $a_e = 1.15965218059(13) \times 10^{-3}$ — thirteen significant figures — and the calculated value, worked out through five orders in α (over twelve thousand Feynman diagrams at the fifth), agrees within the combined uncertainties, at the level of about one part in 10^{12} .

That the series diverges is not a suspicion but an argument, due to Dyson in 1952, and it is worth sketching because it is pure \$3 reasoning. If the series in α had any nonzero radius of convergence, it would also converge for small *negative* α , since a radius means a disc. But with $\alpha < 0$ like charges attract: the vacuum could lower its energy without bound by spontaneously separating clumps of like charge, so there is no stable ground state and no sensible theory at all. A function cannot be analytic at a point on the boundary between "fine" and "catastrophically ill-defined". Hence the radius is zero. (The mechanism, if you want one: the number of Feynman diagrams at order n grows like $n!$, exactly as in our toy integral.)

So how is anyone getting twelve digits? By the grind box. With $x = \alpha/\pi \approx 1/430$, optimal truncation sits near $N_* \approx 430$ terms and the accuracy floor is $e^{-\pi/\alpha} \sim 10^{-187}$. Nobody is anywhere near the turning point; the terms are still falling steeply at fifth order and will keep falling for another four hundred. **Physicists are not being sloppy with a divergent series. They are using an asymptotic expansion correctly, in the regime where the error bound is the first omitted term and that term is astronomically small.** Chapter 5.9 makes this precise and shows what the leftover 10^{-187} actually is.

5 · Dimensional analysis and orders of magnitude

The last tool in this chapter costs almost nothing and repeatedly returns answers that look like they should have required a calculation.

Homogeneity, and why it is not a convention

Every additive term in a physical equation must have the same dimensions. The reason is not etiquette: a physical law must hold whatever units you measure in. Suppose an equation reads $A = B + C$ where B is a length and C a time. Switch from metres to feet and B changes by a factor of 3.28 while C does not; an equation true in one unit system would be false in another, which would make the choice of units a piece of physics. It is not. Hence $[B] = [C]$, always. In mechanics we track three independent dimensions — mass M , length L , time T — and write $[v] = LT^{-1}$, $[F] = MLT^{-2}$, and so on.

The same argument in its stronger form is the **Buckingham π idea**: if a physical relation involves n dimensional quantities built from k independent dimensions, it is equivalent to a relation among just $n - k$ dimensionless combinations. The reason is simple enough to state in one sentence: you may use k of the quantities to *define* your units, setting them to 1, after which only $n - k$ genuinely independent numbers remain to be related. Everything below is that sentence applied.

Demonstration: the pendulum

What can the period T of a pendulum depend on? Plausibly the length L (L), the bob mass m (M), and gravity g (LT^{-2}). That is $n = 4$ quantities and $k = 3$ dimensions, so there is $4 - 3 = 1$ dimensionless group and the physics must be that this group is a constant.

First, mass is the only quantity carrying M , so no dimensionless group can contain it — any power of m would leave an uncancelled M . Conclusion, before any calculation: *the period does not depend on the mass*. Then seek $T \propto L^a g^b$:

$$\begin{aligned} T &= L^a (LT^{-2})^b = L^{a+b} T^{-2b} \\ \implies -2b &= 1, \quad a + b = 0 \implies b = -\frac{1}{2}, \quad a = \frac{1}{2}, \end{aligned} \tag{0.3.25}$$

so $T = C\sqrt{L/g}$ for some pure number C . Two lines, no differential equation, and we have the $\sqrt{L}$ scaling and the independence from mass.

What dimensional analysis cannot give you is C . It happens to be 2π , and getting it requires actually solving the equation of motion — which we do in Worked Example 1. Worse, dimensional analysis is *blind to dimensionless parameters*: the amplitude θ_0 is already a pure number, so nothing forbids C from being a function $C(\theta_0)$. It is. The period of a real pendulum grows with amplitude, and the "constant" C is only constant in the small-angle limit. The method tells you the structure and is silent about everything dimensionless — which is precisely why $\alpha \approx 1/137$ had to be measured rather than derived.

The payoff: the Planck scale

Now something that looks like it should be impossible. Take the three constants that define the three great theories:

CONSTANT	BELONGS TO	DIMENSIONS
$c = 3.00 \times 10^8 \text{ m s}^{-1}$	Relativity	L T^{-1}
$\hbar = 1.055 \times 10^{-34} \text{ J s}$	Quantum mechanics	$\text{M L}^2 \text{T}^{-1}$
$G = 6.674 \times 10^{-11} \text{ m}^3 \text{kg}^{-1} \text{s}^{-2}$	Gravity	$\text{M}^{-1} \text{L}^3 \text{T}^{-2}$

Ask for a length: $\ell = G^a \hbar^b c^d$. Matching exponents of M, L, T gives three linear equations:

$$\text{M} : -a + b = 0, \quad \text{L} : 3a + 2b + d = 1, \quad \text{T} : -2a - b - d = 0. \quad (0.3.26)$$

From the first $b = a$; the third then gives $d = -3a$; substituting into the second, $3a + 2a - 3a = 2a = 1$, so $a = b = \frac{1}{2}$ and $d = -\frac{3}{2}$:

$$\begin{aligned} \ell_P &= \sqrt{\frac{\hbar G}{c^3}} = 1.616 \times 10^{-35} \text{ m}, \\ t_P &= \frac{\ell_P}{c} = \sqrt{\frac{\hbar G}{c^5}} = 5.39 \times 10^{-44} \text{ s}, \\ m_P &= \sqrt{\frac{\hbar c}{G}} = 2.18 \times 10^{-8} \text{ kg}. \end{aligned} \quad (0.3.27)$$

The mass looks unremarkable — about the mass of a flea's egg — until you convert it to an energy: $m_P c^2 = 1.22 \times 10^{19} \text{ GeV}$, roughly 10^{15} times the energy the Large Hadron Collider delivers per collision.

Now the part that matters. The three equations (0.3.26) form a 3×3 linear system whose coefficient matrix has determinant

$$\det \begin{pmatrix} -1 & 1 & 0 \\ 3 & 2 & 1 \\ -2 & -1 & -1 \end{pmatrix} = -1(-2 + 1) - 1(-3 + 2) + 0 = 2 \neq 0. \quad (0.3.28)$$

A nonzero determinant means the system has exactly one solution — for a length, and equally for any other target dimension. So ℓ_P is not *a* length you can build from G , $\hbar$, c ; it is **the only one**. There is no dimensionless knob to slide it, no free parameter, no alternative combination. Combine gravity with quantum mechanics and relativity and a length 10^{-35} m appears whether you wanted it or not.

That is the content of the sentence in Chapter 7.1 that quantum gravity has a built-in scale, and it is worth contrasting with electromagnetism, where the analogous attempt fails. From $\hbar$, c and the Coulomb constant $k_e \equiv e^2/4\pi\epsilon_0$ (dimensions $\text{M L}^3 \text{T}^{-2}$, i.e. energy $\times$ length) you can build no length at all, because those three quantities are dimensionally *dependent*: the combination $k_e/\hbar c = \alpha$ is already a pure number, so the corresponding determinant vanishes. That absence of an intrinsic scale is the deep reason QED behaves so differently from gravity, and it is why the gravitational coupling being dimensionful is exactly the disease diagnosed by power counting in Chapter 7.1.

Orders of magnitude

The habit that goes with dimensional analysis is estimating first and computing later — deciding whether an effect can possibly matter before spending a week on it. One example, using only what is on this page. How does gravity compare to electricity inside an atom? Both forces fall as $1/r^2$, so the ratio is a pure number independent of separation:

$$\frac{F_{\text{grav}}}{F_{\text{elec}}} = \frac{G m_e m_p}{k_e} = \frac{(6.674 \times 10^{-11})(9.11 \times 10^{-31})(1.673 \times 10^{-27})}{2.307 \times 10^{-28}} \approx 4.4 \times 10^{-40}. \quad (0.3.29)$$

Forty orders of magnitude. This is why no chemistry textbook mentions gravity, and — read the other way — why quantum gravity is experimentally out of reach: to make gravity comparable to the other forces you must reach the Planck energy, which is where (0.3.27) said it would be. A single ratio, computed in one line, explains both the structure of Part IV and the predicament of Part VII.

Before spending these tools, one expansion from outside physics, where the mathematics is not analogous but identical.

FAMILIAR GROUND

You have been performing first-order Taylor expansions in clinic for your entire career, under the name "the odds ratio approximates the relative risk when the outcome is rare." Let us actually do the expansion, because it tells you something the rule of thumb does not.

With event probabilities p_1 (treated) and p_2 (control),

$$RR = \frac{p_1}{p_2}, \quad OR = \frac{p_1/(1-p_1)}{p_2/(1-p_2)} = \frac{p_1(1-p_2)}{p_2(1-p_1)} = RR \cdot \frac{1-p_2}{1-p_1}.$$

So the two differ by exactly the factor $(1-p_2)/(1-p_1)$ — no approximation yet. Write $p_1 = R p_2$ with $R = RR$, and expand that factor in the small quantity p_2 using the geometric series $(1-p_1)^{-1} = 1 + p_1 + p_1^2 + \dots$:

$$\frac{OR}{RR} = \frac{1-p_2}{1-Rp_2} = 1 + \frac{p_2(R-1)}{1-Rp_2} = 1 + p_2(R-1) + O(p_2^2).$$

The middle expression is exact; the last is the first-order expansion. Read the leading term: **the fractional discrepancy between OR and RR is approximately (baseline event rate) $\times$ (RR – 1)**. Two consequences that the usual "if the outcome is under 10%" formulation cannot express:

1. The rate is only half the story. The error is controlled by the *product* $p_2(R-1)$, so a large treatment effect degrades the approximation just as fast as a common outcome. With a control rate of 5% — comfortably "rare" — and $RR = 5$, the exact factor is $1 + \frac{0.05 \times 4}{1-0.25} = 1.267$: the odds ratio is 6.33 against a relative risk of 5, an overstatement of 27%. The rule of thumb waves this case through; the expansion does not.

2. It tells you the rate of decay, not just a threshold. At $p_2 = 2\%$ with $RR = 2$ the discrepancy is 2.1%; at 10% it is 12.5%; at 30% it is 75% ($OR = 3.5$ versus $RR = 2$). Notice also that the first-order estimate itself degrades: it predicts 2%, 10% and 30% against true values of 2.1%, 12.5% and 75%. The expansion is warning you, in the only way an expansion can, that by $p_2 = 0.3$ its own higher terms have taken over and the approximation should be abandoned. A good expansion tells you when to stop trusting it — which, as §4 showed, is the same service the first omitted term performs for QED.

The second everyday expansion is (0.3.9). A hazard ratio is reported as $HR = e^\beta$, and $e^\beta \approx 1 + \beta$ is why a log-hazard of -0.05 is read off as "a 5% reduction". It is really $e^{-0.05} = 0.9512$, a 4.88% reduction; the error is the next term, $\beta^2/2 = 0.125\%$. At $\beta = 0.5$ the same reflex gives "50% increase" for a true $HR = 1.6487$ — a 65% increase. Same series, same failure mode, same fix: look at the next term.

△ WHY THIS ISN'T OBVIOUS

You would be forgiven for assuming that a function with infinitely many derivatives is determined by them. It is not, and the counterexample is one you can check by hand.

Define $f(x) = e^{-1/x^2}$ for $x \neq 0$ and $f(0) = 0$. This function is smooth everywhere, including at the origin, where it flattens out so aggressively that it beats every polynomial. Differentiating repeatedly gives $f^{(n)}(x) = P_n(1/x) e^{-1/x^2}$ for some polynomial P_n — true for $n = 0$, and each derivative of such an expression is again of that form by the chain and product rules. To evaluate the derivatives at the origin, use the definition:

$$f^{(n)}(0) = \lim_{h \rightarrow 0} \frac{f^{(n-1)}(h) - 0}{h} = \lim_{u \rightarrow \pm\infty} u P_{n-1}(u) e^{-u^2} = 0,$$

where $u = 1/h$, and the limit vanishes because e^{-u^2} beats any power: from the exponential series with all-positive terms, $e^{u^2} \geq u^{2m}/m!$, so $|u^k e^{-u^2}| \leq m! |u|^{k-2m} \rightarrow 0$ once $2m > k$.

So every Taylor coefficient at the origin is zero. The Taylor series of f is $0 + 0x + 0x^2 + \dots$, which converges beautifully — everywhere, absolutely, to the function 0. And f is not 0: $f(1) = e^{-1} \approx 0.368$. The series converges, and it converges to the wrong function. Recall the warning after (0.3.6): what matters is whether $R_{n+1} \rightarrow 0$, and here $R_{n+1} = f(x)$ for every n , unchanging and never small.

Smoothness is not analyticity.

Now the physics, because this is not a pathology — it is a sector of reality. Perturbation theory produces a power series in a coupling g . Any quantity of the form e^{-c/g^2} has, by the argument above, an identically zero power series at $g = 0$: it is *invisible at every order of perturbation theory*, to all orders, forever. Such terms are exactly what the accuracy floor of §4 was made of — an asymptotic series bottoms out at $e^{-1/x}$ because that is the size of what it cannot see.

And those terms are real. Quantum tunnelling amplitudes go as $e^{-S/\hbar}$; instantons in Yang–Mills theory carry a factor $e^{-8\pi^2/g^2}$; the confinement of quarks and the mass gap of QCD are non-perturbative phenomena of exactly this kind. When Chapter 6.5 says that the proton's mass does not come from the Higgs but from the strong interaction's own dynamics, it is describing physics that lives entirely in the invisible sector defined by e^{-1/x^2} . The most important effects in the strong force are the ones that every term of the series says are zero.

6 · Worked examples

WORKED EXAMPLE 1 — WHY EVERYTHING IS A HARMONIC OSCILLATOR

Take any smooth potential energy $V(x)$ with a minimum at x_0 . What is the motion of a particle released nearby?

Expand V about x_0 using (0.3.6), writing $u = x - x_0$:

$$V(x) = V(x_0) + V'(x_0)u + \frac{1}{2}V''(x_0)u^2 + \frac{1}{6}V'''(x_0)u^3 + \dots$$

Now kill the terms one at a time.

The constant is irrelevant. Only differences of potential energy have physical consequences — force is $-V'$, and adding a constant changes no derivative. Drop it.

The linear term vanishes, because it is a minimum. $V'(x_0) = 0$ is the definition of a stationary point. This is the crucial step and it is not an approximation: at an equilibrium the first-order term is *exactly* absent. Equilibrium is precisely the condition that makes the quadratic term leading.

The cubic and beyond are small. The ratio of the cubic to the quadratic term is $\frac{1}{3} |V'''/V''| |u|$, so they are negligible provided the amplitude satisfies $|u| \ll 3 |V''/V'''|$. That is a quantitative criterion, not a hope.

What is left is

$$V(x) \approx \frac{1}{2}k u^2, \quad k \equiv V''(x_0), \quad \text{so} \quad F = -\frac{dV}{du} = -k u,$$

and Newton's second law reads $m\ddot{u} = -ku$. From Chapter 0.1 we know the function that is minus its own second derivative, so $u(t) = A \cos(\omega t + \varphi)$ with

$$\omega = \sqrt{\frac{V''(x_0)}{m}}.$$

Every stable system, close enough to equilibrium, is a harmonic oscillator with spring constant $V''(x_0)$. Not "can be modelled as" — *is*, to leading order, with an error you can bound. The harmonic oscillator is not a special case that physics happens to like; it is the generic case, the first surviving term of the expansion of anything stable.

Concretely: the pendulum. A bob on a rigid rod of length L has height $-L \cos \theta$ below the pivot, so $V(\theta) = -mgL \cos \theta$ and the kinetic energy is $\frac{1}{2}mL^2\dot{\theta}^2$ — the effective mass for the coordinate θ is mL^2 . Then $V' = mgL \sin \theta$, which vanishes at $\theta = 0$ as required, and $V'' = mgL \cos \theta \rightarrow mgL$. So

$$\omega = \sqrt{\frac{mgL}{mL^2}} = \sqrt{\frac{g}{L}} \quad \implies \quad T = \frac{2\pi}{\omega} = 2\pi \sqrt{\frac{L}{g}}.$$

There is the 2π that §5 said dimensional analysis could not supply, and the mass has cancelled exactly as dimensional analysis promised it must. Equivalently, expanding the cosine with (0.3.8), $V =$

$-mgL \left(1 - \frac{\theta^2}{2} + \frac{\theta^4}{24} - \dots\right)$: constant, then the harmonic term $\frac{1}{2}mgL\theta^2$, then a quartic correction whose effect is to make the period depend on amplitude at relative order θ_0^2 . The expansion cannot give that coefficient without the perturbation methods of Chapter 0.8, but it already tells you the *form* — the correction is quadratic in amplitude, which is why a pendulum clock keeps time provided you keep the swing small.

Where this goes. This one paragraph is why the harmonic oscillator will not leave you alone: Chapter 0.8 (normal modes of coupled systems), Chapter 4.5 (the quantum oscillator and its ladder operators), Chapter 5.3 (a free quantum field is an infinite collection of oscillators, one per momentum mode — which is where particles come from), Chapter 7.4 (the string's vibrational modes, which is where the graviton comes from). Quantum field theory is built out of oscillators for exactly the reason on this page: expand any stable field configuration about its minimum and the quadratic term is what you get first.

WORKED EXAMPLE 2 — THE NEXT RELATIVISTIC CORRECTION, AND WHERE IT IS SEEN

Chapter 0.1 expanded $E = \gamma mc^2$ far enough to find $\frac{1}{2}mv^2$. With the binomial series (0.3.12) we can now go as far as we like at no extra cost. Put $x = v^2/c^2$ into $(1 - x)^{-1/2}$:

$$\gamma = 1 + \frac{1}{2} \frac{v^2}{c^2} + \frac{3}{8} \frac{v^4}{c^4} + \frac{5}{16} \frac{v^6}{c^6} + \dots$$

$$E = \gamma mc^2 = \underbrace{mc^2}_{\text{rest}} + \underbrace{\frac{1}{2}mv^2}_{\text{Newton}} + \underbrace{\frac{3}{8} \frac{mv^4}{c^2}}_{\text{first correction}} + \dots$$

How good is the truncation? At $v = 0.1c$ the exact γ is 1.0050378153, while $1 + \frac{1}{2}x = 1.0050000000$ and $1 + \frac{1}{2}x + \frac{3}{8}x^2 = 1.0050375000$. The error after two terms is 3.15×10^{-7} , and the first omitted term is $\frac{5}{16}x^3 = 3.13 \times 10^{-7}$ — agreement to two figures, exactly as the remainder theory said. Relative to the kinetic energy the correction is $\frac{3}{4}v^2/c^2$, i.e. 0.75% at a tenth the speed of light.

A sign that catches people out. Quantum mechanics works with momentum, not velocity, and the relativistic momentum is $p = \gamma mv$, not mv . Expanding the energy in p instead requires $\sqrt{1 + x}$ rather than $(1 - x)^{-1/2}$:

$$E = \sqrt{p^2 c^2 + m^2 c^4} = mc^2 \sqrt{1 + \frac{p^2}{m^2 c^2}} = mc^2 + \frac{p^2}{2m} - \frac{p^4}{8m^3 c^2} + \dots$$

The correction is now *negative*. Both results are correct; they disagree because v and p are not proportional once $\gamma \neq 1$, so "fourth order in v " and "fourth order in p " are different expansions. This is a standing hazard with series: *the coefficients depend on what you chose to expand in*, and only the physical answer is invariant.

Where the term is seen. The Hamiltonian correction $-p^4/8m^3 c^2$ is the leading relativistic contribution to atomic energy levels. In hydrogen the electron's typical speed is $v \approx \alpha c$ (Chapter 4.5), so this term is smaller than the Rydberg energy by a factor $\sim \alpha^2 \approx 5 \times 10^{-5}$ — shifts of order 10^{-4} eV, which together with the spin-orbit and Darwin terms make up the **fine structure** of the hydrogen spectrum. It was measured in the nineteenth century, before anyone knew what it was. It is the third term of a Taylor series.

7 · Your turn

PROBLEM 1 — RADIUS OF CONVERGENCE, TWICE

(a) Find the radius of convergence of the Maclaurin series of $f(x) = \frac{1}{4+x^2}$, first by writing the series explicitly and applying the ratio test, then explain the answer geometrically.

(b) Find the radius of convergence of $\sum_{k=1}^{\infty} \frac{x^k}{k 3^k}$, identify the function it represents, and say what happens at the two endpoints $x = \pm 3$.

– Solution

(a) Factor out the 4 and use the geometric series with $u = -x^2/4$:

$$\frac{1}{4+x^2} = \frac{1}{4} \cdot \frac{1}{1+x^2/4} = \frac{1}{4} \sum_{k=0}^{\infty} \left(-\frac{x^2}{4}\right)^k = \frac{1}{4} - \frac{x^2}{16} + \frac{x^4}{64} - \dots$$

Successive terms are in the ratio $|-x^2/4| = x^2/4$, so the series converges when $x^2/4 < 1$, i.e. $|x| < 2$: the radius is $R = 2$.

Geometrically: as a function of a complex variable, $1/(4+z^2)$ blows up where $z^2 = -4$, i.e. at $z = \pm 2i$ — both at distance 2 from the origin. The disc of convergence is the largest disc centred at 0 containing no singularity, and those two poles stop it at radius 2. On the real line the function is smooth and bounded everywhere, with a maximum of $1/4$ at the origin and no feature at $x = \pm 2$; the boundary is invisible from the real axis, as in §3.

(b) Coefficients $a_k = 1/(k 3^k)$, so

$$\left| \frac{a_k}{a_{k+1}} \right| = \frac{(k+1)3^{k+1}}{k 3^k} = 3 \cdot \frac{k+1}{k} \rightarrow 3,$$

hence $R = 3$. The function: from (0.3.9) with $x \rightarrow -x/3$,

$$\sum_{k \geq 1} \frac{x^k}{k 3^k} = \sum_{k \geq 1} \frac{(x/3)^k}{k} = -\ln\left(1 - \frac{x}{3}\right),$$

which indeed has a singularity at $x = 3$, on the real axis this time — no complex detective work needed.

Endpoints. The ratio test says nothing when the limit equals 1, and here the two ends genuinely differ. At $x = 3$ the series is $\sum 1/k$, the harmonic series, which diverges — matching the fact that $-\ln(1 - x/3) \rightarrow +\infty$ there. At $x = -3$ it is $\sum (-1)^k/k$, the alternating harmonic series, which converges (to

$-\ln 2$), and the function is perfectly finite there. So the boundary circle of a power series can behave differently at different points on it; the radius tells you about the interior only.

PROBLEM 2 — THE LORENTZ FACTOR TO FOURTH ORDER

Expand $\gamma = (1 - v^2/c^2)^{-1/2}$ to order v^4 from the binomial series. Evaluate the truncation error at $v = 0.1c$ and check it against the first omitted term. At what speed does the v^4 term contribute 1% of the kinetic energy, and where does this term show up physically?

– Solution

With $\alpha = -\frac{1}{2}$ and argument $-v^2/c^2$, the coefficients from (0.3.11) are $\binom{-1/2}{1} = -\frac{1}{2}$, $\binom{-1/2}{2} = \frac{(-1/2)(-3/2)}{2} = \frac{3}{8}$, $\binom{-1/2}{3} = \frac{(-1/2)(-3/2)(-5/2)}{6} = -\frac{5}{16}$. Each is multiplied by $(-v^2/c^2)^k$, so all signs come out positive:

$$\gamma = 1 + \frac{v^2}{2c^2} + \frac{3v^4}{8c^4} + \frac{5v^6}{16c^6} + \dots$$

Numerical check at $v = 0.1c$. Exact: $\gamma = 1/\sqrt{0.99} = 1.0050378153$. Truncated at v^4 : $1 + 0.005 + 0.0000375 = 1.0050375000$. Error = 3.153×10^{-7} . First omitted term: $\frac{5}{16}(0.1)^6 = 3.125 \times 10^{-7}$. The two agree to better than 1% of themselves — the remainder really is "the next term, near enough".

The 1% speed. Kinetic energy is $T = (\gamma - 1)mc^2 = \frac{1}{2}mv^2 + \frac{3}{8}mv^4/c^2 + \dots$, so the ratio of the second term to the first is

$$\frac{\frac{3}{8}mv^4/c^2}{\frac{1}{2}mv^2} = \frac{3}{4} \frac{v^2}{c^2} = 0.01 \implies \frac{v}{c} = \sqrt{\frac{0.04}{3}} = 0.1155.$$

So Newtonian kinetic energy is good to 1% up to about 11.5% of light speed — $3.5 \times 10^7 \text{ m s}^{-1}$, which is faster than anything in the solar system and slower than everything in a particle accelerator. That single number explains why relativity went unnoticed until people started looking at light and electrons.

Physically. Rewritten in momentum, the same expansion gives the correction $-p^4/8m^3c^2$ to the Hamiltonian (Worked Example 2), which is the leading relativistic contribution to atomic fine structure — of relative size $\alpha^2 \approx 5 \times 10^{-5}$ in hydrogen.

PROBLEM 3 — HOW FAR CAN YOU PUSH THE RARE-OUTCOME APPROXIMATION?

A trial reports 2-year recurrence in 25% of controls and 15% on treatment. Compute the relative risk and the odds ratio exactly, express the discrepancy as a percentage, and compare it with both the exact factor $1 + \frac{p_2(R-1)}{1-Rp_2}$ and the first-order estimate $1 + p_2(R-1)$ from the Familiar Ground box. What does the size of the second-order term tell you about reporting the odds ratio as though it were a risk ratio here?

– Solution

With $p_1 = 0.15$ and $p_2 = 0.25$:

$$RR = \frac{0.15}{0.25} = 0.600, \quad OR = \frac{0.15/0.85}{0.25/0.75} = \frac{0.176471}{0.333333} = 0.529412.$$

The ratio is $OR/RR = 0.882353$, i.e. the odds ratio sits **11.76%** below the relative risk.

Against the exact formula, with $R = 0.6$ and $p_2 = 0.25$ (so $Rp_2 = p_1 = 0.15$):

$$1 + \frac{p_2(R-1)}{1-Rp_2} = 1 + \frac{0.25 \times (-0.4)}{0.85} = 1 - \frac{0.1}{0.85} = 0.882353. \checkmark$$

Exact, as it must be — that expression involved no approximation.

Against the first-order estimate: $1 + p_2(R-1) = 1 - 0.100 = 0.900$, predicting a 10.0% discrepancy where the truth is 11.76%. The expansion of $1/(1-0.15) = 1 + 0.15 + 0.0225 + \dots$ shows where the missing 1.76 points come from: the second-order term is 15% of the first, which is exactly p_1 .

Interpretation. Reported as an odds ratio, this treatment "reduces the odds by 47%"; the actual reduction in risk is 40%. At a 25% baseline the odds ratio is not a stand-in for the risk ratio — it exaggerates the effect by a fifth of the effect size, and the exaggeration is always away from 1. The expansion also tells you the fix without needing a new rule of thumb: the discrepancy scales as $p_2(R-1)$, so if you want it under 5% with $R = 0.6$ you need $p_2 \lesssim 0.12$. Note finally that the first-order estimate was itself 15% off — when the leading correction is 12%, the next one is around 1–2%, and you are no longer in the regime where one term is enough.

PROBLEM 4 — THE BOHR RADIUS FROM DIMENSIONS ALONE

(a) The hydrogen atom is built from $\hbar$, the electron mass m_e , and the Coulomb constant $k_e = e^2/4\pi\epsilon_0$ (dimensions of energy $\times$ length). Find the unique length that can be built from them and evaluate it. What else can you get, and what can dimensional analysis *not* tell you?

(b) Show that $\hbar$, c and k_e alone cannot produce a length, and say why that is structurally different from the G , $\hbar$, c case of §5.

– Solution

(a) $[\hbar] = \text{ML}^2\text{T}^{-1}$, $[m_e] = \text{M}$, $[k_e] = \text{ML}^3\text{T}^{-2}$. Seek $a = \hbar^A m_e^B k_e^C$:

$$\text{M} : A + B + C = 0, \quad \text{L} : 2A + 3C = 1, \quad \text{T} : -A - 2C = 0.$$

The third gives $A = -2C$; substituting into the second, $-4C + 3C = -C = 1$, so $C = -1$, $A = 2$, and $B = -(A + C) = -1$. Hence

$$a_0 = \frac{\hbar^2}{m_e k_e} = \frac{(1.0546 \times 10^{-34})^2}{(9.109 \times 10^{-31})(2.307 \times 10^{-28})} = 5.29 \times 10^{-11} \text{ m},$$

which is the Bohr radius, correct including its numerical factor — a piece of luck, not a guarantee. The same three constants give exactly one energy, $k_e/a_0 = m_e k_e^2/\hbar^2 = 27.2 \text{ eV}$, and exactly one speed, $k_e/\hbar = 2.19 \times 10^6 \text{ m s}^{-1}$.

What is fixed and what is not. Fixed: the *combination*, hence all the scaling. Double the electron mass and the atom shrinks by a factor of two; that conclusion needs no quantum mechanics. Not fixed: every pure number. The ground-state energy is -13.6 eV , which is $-\frac{1}{2}$ of the natural energy scale, and dimensional analysis cannot produce that $\frac{1}{2}$ any more than it produced the pendulum's 2π . Nor can it produce the spectrum's $1/n^2$, since n is dimensionless. Notice also the third result: dividing the natural speed by c gives $k_e/\hbar c = \alpha = 1/137$, the electron's typical speed in units of light speed — which is exactly the small parameter that made the relativistic correction of Worked Example 2 of relative size α^2 .

(b) Try $\text{L} = \hbar^A c^B k_e^C$. The exponent matrix, with columns $(\hbar, c, k_e)$ and rows $(\text{M}, \text{L}, \text{T})$, is

$$\begin{pmatrix} 1 & 0 & 1 \\ 2 & 1 & 3 \\ -1 & -1 & -2 \end{pmatrix}, \quad \det = 1(-2 + 3) - 0 + 1(-2 + 1) = 1 - 1 = 0.$$

The determinant vanishes, so the system is singular: either no solution or infinitely many, and never a unique one. The kernel is the reason — the combination $k_e/\hbar c$ has zero dimensions, being the fine structure constant α , so you may multiply any candidate length by any power of α and get another candidate. Electromagnetism, quantum mechanics and relativity together define no length; you must import a mass, such as m_e , before the atom has a size.

The contrast. For $G, \hbar, c$ the determinant was $2 \neq 0$ ((0.3.28)): the three constants are dimensionally independent, they fix a complete system of units, and a length 10^{-35} m exists whether or not anyone wants it. Gravity therefore arrives with its own scale built in, and the dimensionful coupling G is exactly what makes the perturbative expansion of gravity break down at that scale — the power counting of Chapter 7.1.

THE BRICK YOU JUST LAID

You have Taylor's theorem *with* its remainder, derived by nothing more exotic than repeated integration by parts, in both integral and Lagrange forms — so that every truncation you ever make comes with a bound rather than a hope. You have the handful of expansions that carry most of physics, including the binomial series for arbitrary real exponent, and you have Euler's formula as a consequence of the exponential series rather than as a slogan: $e^{i\theta}$ is a rotation, which is why quantum mechanics conserves probability while first-order kinetics does not conserve drug. You know that the radius of convergence is set in the complex plane even for a real function that never misbehaves; that most series in physics diverge and are used anyway, correctly, by truncating near the smallest term; that smooth is not analytic and that an entire sector of physics hides in the difference; and that dimensional analysis gets you the structure of an answer for free while remaining permanently blind to pure numbers.

Where this gets spent. *Taylor expansion* → Chapter 1.2 (varying an action is expanding it to first order in a whole function), Chapter 2.5 (the Newtonian limit of relativistic dynamics), Chapter 5.6 (every Feynman diagram is a term in a series). *Euler's formula* → Chapters 0.8 and 0.9 (oscillators, Fourier analysis) and then the whole of quantum mechanics from Chapter 4.2 onward. *Asymptotic series* → Chapter 5.9, where the divergence becomes the renormalization group. *Non-analyticity* → Chapter 6.5, where instantons and confinement live in the part of the answer perturbation theory cannot see. *Dimensional analysis* → Chapter 7.1, where $\ell_P = \sqrt{\hbar G/c^3}$ is the reason quantum gravity is hard rather than merely unfinished. And *small oscillations* → Chapters 0.8, 4.5, 5.3 and 7.4, because "expand about the minimum and keep the quadratic term" is, in the end, why quantum field theory is built out of harmonic oscillators.